THE IMPACT OF DIFFERENT PRODUCTION STYLES OF VIDEOS IN ONLINE EDUCATION

G. Salcas Olguin, I. Alves de Souza, T. Tarraf Varella, A.C. Seabra

Escola Politécnica, University of São Paulo (BRAZIL)

Abstract

At the current given scenario of multiplying educational methods, Massive Open Online Course (MOOCs) in higher education became a trend for many reasons. The MOOCs are a user-friendly learning tool that can be accessed for anyone at anytime. In that way you can spread knowledge to all, increasing the learning opportunities. One resource used in MOOCs is the video class. In most of these videos the professors present key concepts of a particular field. These materials are not so easy to produce and require time and computer skills. Nevertheless, when the video is recorded, students are able to learn at their own pace, repeating the most challenging steps until it is completely understood. Surprisingly, the technique may sometimes make learning a more personal issue as the professor is looking at the student while talking, not to mention the opportunity to use animations or make long and tiresome calculations appear instantly. This work presents a literature review of four different production styles of digital lessons: live classroom recording, shots at an office desk, slideshows and digital drawings. The main objective was to present pros and cons of each production style in the following topics: by ease of creation and editing, efficiency at showing information, and engagement of the viewer. After comparing different approaches in video making for educational purposes, it was possible to identify a new trend based on a mix of video resources and collaborative work. In this new trend, the student has access to a more engaging learning experience and acquire other skills such as teamwork, communication and initiative to look for new information by themselves. In that case the video can be linked with other activities such as forum's, collaborative tasks, quizzes and essays.

Keywords: MOOC, Modern Educational Techniques, videos, Production Style.

1 INTRODUCTION

As reported by Chapman and Gregory (2012), students have 'designers brains'. With that, they meant to synthesize the past studies pointing that each student best acquire knowledge in a different way, a different place or a different time. When we introduce multimedia resources into the learning process, we improve the diversity and give more opportunities for different kinds of learners.

The MOOCs - Massive Open Online Courses, are a new trend in higher education. Colleges around the world strive to design, record, edit and deliver digital courses and thus achieve large number of students.

This new form of education gained ground for its flexibility and accessibility (Arimoto, Barroca and Barbosa, 2014), which is positive for reaching different learning styles and, in some cases, was considered more effective than theoretical texts (Donkor, 2010).

Some issues still require further investigation, especially a better understanding on student engagement and effective learning behind the digital classroom, since these points are critical to validate the continuity of use of the digital teaching.

The aim of this article is to present information on positive aspects of digital courses, especially about the appropriate types of videos, to encourage teachers to use this new tool.

The most common formats of video classes are: captured live classroom, shots at an office desk, slideshows and digital drawings. In the present study, we sought to study these kinds, to analyze their differences and to get data on student engagement in order to choose the most appropriate models depending on the digital course in question. They were analyzed by ease of creation and editing, efficiency at showing information, and engagement of the viewer.

ISBN: 978-84-608-2657-6

2 METHODOLOGY

For this research we used the method of case study research (Yin, 2010). The case study was chosen because it is an empirical research that analyses a contemporary phenomenon within its real-life context. This choice is also due to the fact that research does not have control over behavioural events, dealing with recent situation in the twenty-first century and aiming to understand "how" certain phenomena occur.

In the case study the researcher's goal is to expand and generalize theories (analytical generalization) and not number frequency (statistical generalization) (Lipset, Trow, & Coleman, 1956). The case study is not a data collection tactic or a feature of planning, but a comprehensive research strategy (Stoecker, 1991) with theoretical propositions, a research project model, information sources for the identification and analysis method.

The research project to Yin (2012) should incorporate a theory, that is, the full course of the research requires theoretical propositions providing a direction for selection and data collection and data analysis strategies. Yin sets out four main types of theories: individual (focused on one person as in personality research, behaviour, perception and cognitive development), group (which deals with the interaction between people and in family functioning investigations, work teams and supervision employees), organizational (studying institutions in the structural aspect, bureaucratic, international behaviour and organizational efficiency) and social (addressing cultural, political and economic). In the case of this study, which analyses the progress and effectiveness of the concept of video lessons, it was selected social theory mode.

Besides the definition of a type of theory, the case study needs a project template, which can be a combination of two characteristics: number of case studies and analysis units. A research project can only have one case study under investigation, or multiple case studies, with more than one subject matter. The advantage of multiple case design is that the evidence resulting from research can provide a comprehensive study of an issue, bringing more strength to the arguments used (Herriott & Firestone, 1983). The disadvantage is the time taken for the realization of a multi-case design.

Another feature to be determined on a case study project is the amount of units of analysis. The choice of more than one unit of analysis occurs when within a single unit it is paid more attention to subunits. An object in this situation can be subdivided into other elements of analysis since its composition continues interconnected. This type of project, called embedded case study design, analyses separately each subunit selected. However, if the intention is a more comprehensive examination of an object of study, it is used a holistic project. The holistic design is advantageous when it is impossible to identify a subunit and when the chosen theory itself is holistic in nature. The downside is that in the holistic project the researcher can fail to give necessary attention to a specific phenomenon, which could be treated in a subunit of an embedded project.

Since different types of video lessons will be studied, taking into account social issues involving political, historical, economic and cultural aspects worldwide, it was chosen holistic design of multiple cases. To conduct the case study we used information from articles, dissertations, thesis and journals.

According to Richard Yin (2010), within the main specific methods of analysis for case study are the suitability to the standard (which compares a fundamental empirical pattern with another base prognostic), building the explanation (for a phenomenon that stipulates a set of links caused by it), the time series analysis (historical analysis of a phenomenon) and logical models of program (combination of techniques to the pattern adequacy to analysis of time series, which is a historical tracking of a phenomenon that by turn, is analysed with an empirical standard and prognostic basis). As this research uses social theories and has a well-established historical focus, it makes sense to opt for a method involving the historical development of a phenomenon. The complexity of the matter does not allow identifying simple causal relationships, only an understanding and interpretation of the phenomena of interest. Therefore to "fit the pattern" is the most appropriate method of analysis. In this methodology, when the analysis of results from data collect match the patterns found in similar research, it is reinforced the internal validity of case study, allowing to infer a solid conclusion on the observed phenomena.

3 DISCUSSION

For historical analysis, were selected four categories that depict a particular time:

Classroom Videos: First video mode, where lessons of professors in the classroom are lively recorded.

Recording Studio: The video lessons are recorded with a script in different environments, and require later editing.

Digital Drawing: the use of specific programming features to demonstrate concepts in videos

Animations: similar statements as a virtual laboratory where the video demonstrates simulations.

It was not purpose of this article to establish these four categories in order to categorize all types of existing videos. It was meant to point the most eligible elements in the scientific community for education issues that bring impact on the current classes. In the following section, the four categories will be presented.

3.1 Classroom Videos

It is common sense that digital technology provides new ways – easier ones – of doing old activities. Education is included on it, as proved by Feynman's videos, back to the lectures at the Cornell University in 1964. Those videos were recovered and have been available to the public since 2009 by Bill Gates at Tuva Project [17].

They consist of regular lectures fully recorded, that is, a video of the class the way it was exposed lively to the students. At that time, the obstacle was the distribution of the material, once was the first videocassette recorder launched by Sony in 1964 [18], so it could not enlarge the viewers at that moment.

The pioneering of these famous videos show that no advanced technology is necessary to open spread knowledge, learning may be achieved by simply recording regular classes, and it even does not have to be available online. Khan, in his book Web-Based Training [8] defines classroom videos as part of the more general unit called Presentation, or a kind of theory web-based learning, which basically contains definitions and facts. Videoconferences are also put into this class.

The recorded lessons enhances flexibility, by letting students watch the lectures on favourable time, at a personalized speed – faster or slower, and even go back to specific parts that could not be understood by the first time of watching.

Besides, the availability of this material is not conflicting with the attendance of regular classes; it only provides for the students another method of watching them. In 2008, Cardall, Krupat and Ulrich confirmed this fact through an inquiry at Harvard Medical Schools, as they concluded that the majority of the students remained attending live classes even after the videos were available.

Even if the main trend remains on live attendance, the students also review lessons, which show that the classroom type of videos is effective for later revision of the class. Evans, 2008, concluded that they are even better than written notes.

In addition, recorded lessons provide data for helping professor to improve his teaching techniques. Nowadays, even glasses with embedded cameras can be used to record lessons, as experienced by Odhabi and Nicks-McCaleb in 2011. Furthermore these videos can be easily integrated to web-based learning tools.

The digital education inclusion permitted is a positive side of the recorded classrooms, but it must have a feedback of the students. This kind of web-based learning has to be improved to go beyond an information repository. As Pirie stated in 1996, it has to be able to teach and not only to inform. Otherwise it would only be a replacement of a newspaper on academic field.

Nowadays, it is possible for the classrooms videos to be played simultaneously to its recording, what permits the students to interact alive. The videoconferences are a tool for it and enrich the process.

In resume, the classroom recording videos are an effective alternative method for watching the regular lessons, that is, the pedagogical manner of teaching remains the same, once it is just a reproduction of the class. The positive point is the flexibility to the way of attending classes due to its remote characteristic.

3.2 Studio Recording

The studio recording consists of a lecture recorded at specific environment other than the classroom. There is no characteristic length for the lecture, depending on the subject. For better understanding, it might be taken as basis the example of "Telecurso", a project developed by private and public institutions in Brazil, Fundacao Roberto Marinho and FIESP - Fundacao das Industrias do Estado de Sao Paulo (http://educacao.globo.com/telecurso/).

The lectures made for the project consist of classes recorded in a studio or in many different places depending on the aim of the class. With a more professionalizing character, the lectures were made so that the students were closer to the practice than conventional classes or classroom recordings.

"Telecurso" certifies a student with a high school degree or other qualifications, as technical degree, for example. The staff is composed by tutors (qualified people who had specific times to clear doubts) and there are exams in order to select who is able to receive the certification. It is well known for the massification of the learning process, which is good for opening the education and the opportunities for many people who hadn't had the opportunity before, but there were some failures as lack of individuality with the students in the learning process.

A studio recording does not need to be independent of the conventional classes. It is a strong tool to be used when necessary, since there is the possibility of transporting students virtually to different environments. It is somehow better for some students to keep the class more interactive and attractive than a classroom recording because of the possibilities that it brings.

An example of that is when a high school teacher uses the videos of Biology from "Telecurso" to show real examples of the subject or when chemistry teachers use the laboratory recordings when a real one is not available. Even if a real laboratory is available for a class, it is important for the student to know about the videos so he can study through it whenever he wants to.

The digital class length usually is shorter than a normal class because the time can be more optimized. Students have the opportunity of returning and watching the video again at difficult parts and as the class is not interrupted for any reason, the same subject taught in the classroom takes more time than in a studio recording.

It is important to notice that the oral modes are different from normal classroom videos. One difficulty is to keep the attention of the students, so it is important not to speak too fast or too slow and try to use an informal vocabulary.

3.3 Digital Drawing

Nowadays, the symbol of web-based learning is a video showing a virtual board and the theory being drawn at it. Khan Academy made them famous by its step-by-step solution and instituted the drawing on the screen with a synchronous narration as a virtual lesson. The *khan style* can be referred to as digital drawing, virtual board or screencasts.

Besides its first focus on high school learning, this technique could be used in college education as well. This style is defined as Demonstration and Drill and Practice, due to manner of showing how the subject works and doing repetitive practice of it.

These videos simulate a chalkboard with the convenience of mobility and interactivity with videos, images and texts, besides the diversity of writing options. Relations with other medias help the understanding, and even illustrate the process being taught. It also stimulates the students that learn better by visualization of the subject.

There is the worry of how much innovation this type of video brings, once it seems to continue the old technology of a board. However, as they are usually made for explanations on mathematics and physics, the resolutions of exercises are precise and easier to understand, since it seems as the narrator is by your side, on a notebook, doing the exercise with you. That is, the virtual board lessons are more personal, and still carry the possibility of reviewing and slowing down the video.

Researchers, from the Open University at UK in 2012, found out that the combining method of writing and speech is one of the strengths of screencasts. And so these virtual lessons are spreading and growing in popularity as their effectiveness is proved and new programs are getting available to ease the making of the videos. Colace, De Santo and Vento evaluated some platforms for this objective in 2003.

The virtual board videos can even solve the small ratio of staff per student at university as pointed by Robinson in 2008, once they recreate the personal assistance atmosphere. Even the students can feel the benefits of this type of remote learning, as stated by Yoon and Sneddon in 2011, as the videos entertain them by the reproduction on devices, such as tablets and cell phones, once used for recreation only.

At last, the screencasts videos provide a close monitoring on the studies and, by its communication through multiple channels, attracts the students and retains their attention.

3.4 Animations

The Applets and animations are recent technologies that open doors to new methods of the learning process. It is largely used at software's "Help files" and can be used to illustrate concepts and interact with the user. The most used platform for developing applets is Java, which means that the only prerequisite is a computer with the common Java platform and a browser, making the Applets an easily accessible tool. The development of new simulations and animations are making easier with experience due to the new libraries and collaborations being created.

A well-known case is the Geogebra software, a program that enables the visualization of many geometric and even algebraic concepts. For example, it is possible to give the notion of limit at a Calculus course and enable the user to choose which function he will study.

Applets are not usually for the whole class, although it is possible to teach a subject by it. The problem of using only applets is that it is usually easier to record some parts and to integrate into it, because it is better to give the same information with simple speeches than make applets the whole class.

There are two main types of applets that differ a lot from each other, the interactive and the non-interactive applets. The non-interactive ones are easily to integrate with videos and conventional classes because they can be used as examples of what was being described. And the interactive ones are more difficult to integrate with the classes but it has been shown that the results are better when evaluating its learning efficiency and the satisfaction of the students. One easy way of interacting is making questions through the video.

It is shown that the process of learning is facilitated when using visual resources. Again, the laboratory example is a good situation where applets are largely used when there is no laboratory available, although it is not replaceable: the difference consists on the difference of practical situations and simulations. There are some examples, though, that applets bring more information than the practical laboratory, as the case of line forces at physics. Even in this case, it is important to have the practical laboratory so that the students can see the real life situations.

All the above data are summarized in the following table:

Table 1: information about the analysis in issue.

Summarized analysis			
	Ease of creation and editing	Efficiency at showing information	Engagement of the viewer
Classroom recording	+++	++	+
Studio recording	++	+	++
Digital Drawing	+	+++	+++
Animations	+	++	+++

Where "+++" indicates a relevant contribution to the parameter in issue, and "+" indicates some failure at it.

4 CONCLUSION

Overtime, more projects of digital lessons of various formats were made, and thus increasing the data basis for new courses, facilitating its making of. Professors, nowadays, can access video resources of other courses and even similar courses made by different institutions, that is, they can learn new methods of teaching and even check the way others are teaching the same subject.

Besides, professors are able to better understand how to make a digital course and, as this subject is trendy now, more research is done about it, getting data about what is effective for learning. The students’ feedback is a relevant index of successful methods at live classrooms, and keeps its value for rating videos as well. The researches done have collected this information and analyzed, making it available for the beginners.

In the path of easing the video making, there is greater availability of animation and interactivity resources now than in old times. The new tools and technologies are easy to be accessed and to be used, overcoming the previous problems. And, in addition, this type of video attracts viewers by its modern appealing.

The cases studied are shown in an historical order, correlated with a difficulty one. By the time editing videos was a hard to find technology, professors started recording its own lessons. With the facilitation of the editing process, it was possible to make studio recordings and make the digital classes a different experience.

New opportunities of videos were able with the birth of the digital drawing, bringing a more proximal experience of, for example, solving exercises. Lastly, the development of Information Technology brought techniques that approximate a real experience, simulating many situations and making them interactivity.

Evaluating methods have still been discussed: due to the big public that digital classes have, it is very difficult to correct a normal exam as in an university or a school. Beyond that, with the conventional method of evaluation, it is not easy to guarantee that it is the user doing the exam or someone else.

To overcome the problem, the nature of the evaluation process is being studied, it is being questioned whether a few theoretical questions concentrated in a few hours really test what a student has learned in a whole semester. There are platforms where the evaluation process is a project to be delivered to many professors, consisting of a new technology for digital learning.

New technologies, however, did not improve only a single type of video classes. For example, it has made possible to perform video conferencing, that is, making the classroom recorded video available lively, even by mobile computers, accessed anywhere with internet connection. The quality of the recorded video also got better, with the auto focus cameras and high-quality images, even at simultaneous transmissions.

For the storage of all this material, there is an increasingly number of web sites, which provide features as video editors, discussion chats for students and tests. These video-based sites have invested in ways to evaluate the viewer's learning, since it is the basis for the continuation of this type of education, ensuring that it is not only informational.

It is important to remark that the digital courses, beyond allowing the enlarging of the public - the greater access generating educational inclusion in digital way, may also broadening the public for future generations, as in the case of Feynman videos.

Each type of video has different characteristics as well as each student has individual preferences. This way, it is important to know many possibilities of making video classes and blend them as necessary, considering the difficulty and the gain of each method in each situation. Mixing the types of videos moderately is very useful to students so that new situations come and improve engagement, as help them to pay attention to the video.

Video experience usually is not enough to be adopted in schools and universities as the whole curriculum, due to difficulties of integrating the evaluating methods and the presence of the professor. As each type of video has different characteristics, live classes have its own characteristics and are still important in the learning process, so that videos are useful to improve the learning experience, not replacing professors.

In resume, the video classes are a method of learning, a new and trendy one, which has its positive and negative points as every other. It has improved by decades, and may be used for direct learning, completion of studies and even for a facilitated review of the live class. As every new method, it has to be taken under vigilance, as it is explored. Thus, it is still a new way of learning available to students, contributing to the flexibility of education and greater ability to customize, once it is up to each one to perform activities on your own time.

ACKNOWLEDGMENTS

This research was supported by Amigos da POLI (http://www.amigosdapoli.com.br). We thank our colleagues from ASK who provided insight and expertise that greatly assisted the research.

REFERENCES

[1] Chapman & Gregory, Differentiated Instructional Strategies: One Size Doesn't Fit All. 2002, p. 19.

[2] Arimoto, Maurício Massaru; Barroca, Leonor and Barbosa, Ellen Francine (2014). Recursos Educa- cionais Abertos: Aspectos de desenvolvimento no cenario brasileiro Open Educational Respources: aspects of development in the Brazilian context. RENOTE - Revista Novas tecnologias da Educação, 12(2).

[3] Donkor, F., The Comparative Instructional Effectiveness of Print-Based and Video-Based Instructional Materials for Teaching Practical Skills at a Distance, 2010.

[4] Yin, R. K. Estudo de caso: planejamento e métodos. Port Alegre: Bookman, 2012.

[5] Lipset, S. M., Trow, M., & Coleman, J. (1956). Union democracy: The inside politcs of the International Typographical Union. New York: Free Press.

[6] Stoecker, R. (1991) Evaluating and rethinking the case study. The Sociological Review, 39, 88 – 112.

[7] Official Richar Feynman Website, available at <http://www.richard-feynman.net/videos.htm>, accessed at 07/sep/15.

[8] Khan, Badrul Huda, Web-based Training. Educational Technology Publications, 2001, 599 pages.

[9] Cardall, Scott; Krupat, Edward; Ulrich, Michael, Live Lecture Versus Video-Recorded Lecture: Are Students Voting With Their Feet? Academic Medicine, Volume 83, Issue 12, December 2008, pages 1174-1178.

[10] Evans, Chris, The effectiveness of m-learning in the form of podcast revision lectures in higher education. Computers & Education, Volume 50, Issue 2, February 2008, pages 491–498.

[11] Odhabi, Hamad; Nicks-McCaleb, Lynn, Video recording lectures: Student and professor perspectives. British Journal of Educational Technology, Volume 42, Issue 2, March 2011, pages 327–336.

[12] Pirie, Susan E. B, Classroom Video-Recording: When, Why and How Does It Ofer a Valuable Data Source for Qualitative Research? October of 1996.

[13] Jordan, C.; Loch, B.; Lowe T.; Mestel B.; Wilkins C., Do short screencasts improve student learning of mathematics? MSOR Connections Vol. 11 No. 4. 2012.

[14] Robinson, Andrew, Easy Implementation of Internet-Based Whiteboard Physics Tutorials. 2008, American Association of Physics Teachers.

[15] Colace, F.; De Santo, M.; Vento, M., "Evaluating on-line learning platforms: a case study," in System Sciences, 2003. Proceedings of the 36th Annual Hawaii International Conference on , vol., no., pp.9 pp.-, 6-9 Jan. 2003.

[16] Yoon, C. & Sneddon, J. (2011). Student Perceptions of Effective Use of Tablet PC Recorded Lectures in Undergraduate Mathematics Courses. International Journal of Mathematical Education in Science and Technology, 42(4), 425-445.

[17] Microsoft Tuva Project Website, available at <http://research.microsoft.com/en-us/research/toys/project-tuva.aspx>, accessed at 19/sep/15.

[18] http://www.sony.net/SonyInfo/CorporateInfo/History/SonyHistory/2-01.html